Lab-8 Ex4 - Evil Numbers

The evil is always in the details. In number theory, an evil number is a non-negative decimal integer that has an even number of 1s in its binary expansion.

For example:

The binary expansion of 3 is **0011**, containing two 1s, so 3 is evil.

The binary expansion of 7 is **0111**, containing three 1s, so 7 is not evil.

A handy conversion table can be found here:

(https://www.rapidtables.com/convert/number/decimal-to-binary.html)

In this exercise, please write a program that requests for two positive integers ***a*** and ***b*** within the range of ***2*** to ***255*** from the user,

and then print all the evil numbers within the range of ***b*** to ***a*** (both inclusive) in descending order. If there are no evil numbers between them, output nothing.

Your program can repeatedly divide the input number by 2 and use the modulus operation to get the successive digits of the number for counting 1s. You will also find that it is very similar to one of the additional examples in the while loop lecture.

You may declare a large enough array to hold the binary digits for easy debugging.

You may assume that the user input is always valid, while ***a*** is always less than or equal to ***b***.

Sample Run #1

23 24

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24
23
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Sample Run #2

34 39

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39
36
34
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